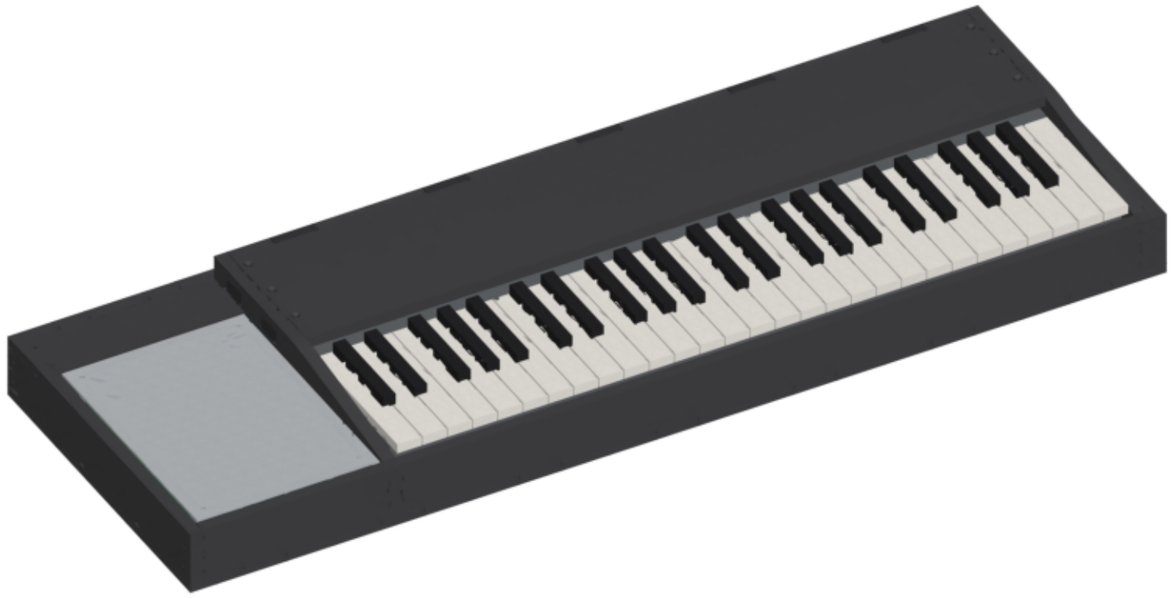
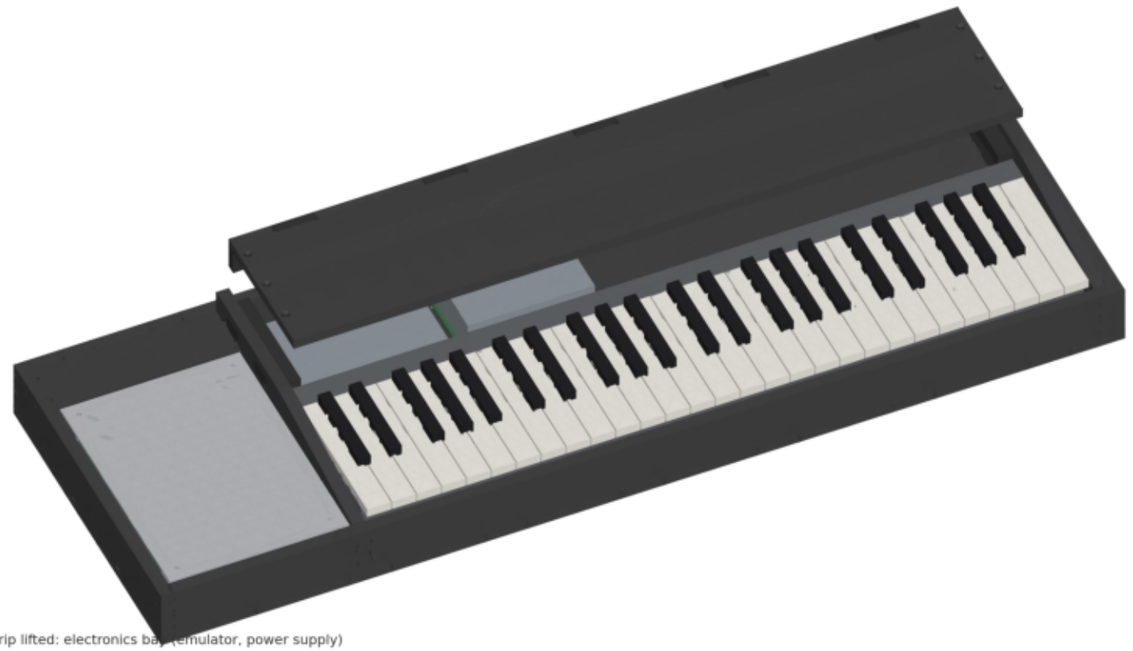


ARP 3620 wooden case Rev0.2 - without lid



strip lifted: electronics bay (emulator, power supply)



rear: connector plate



with lid

Key data

Outside: 911 x 310 mm, height 59 (walls) / 82 (strip) / 95 with lid, plus feet 12

Birch plywood 12 / 9 / 6 mm, covered in black tolex, lid with 4 latches

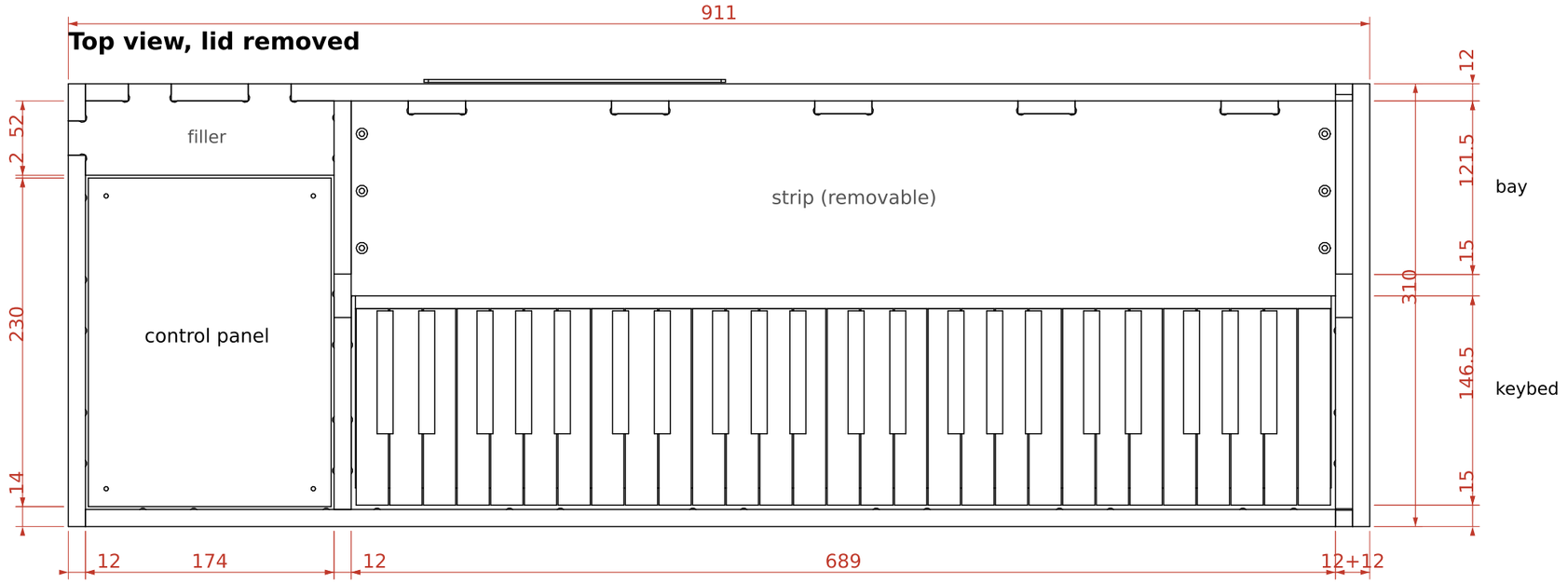
Keybed Fatar TP/9S 49 on two 12 mm risers, control panel 170 x 230 flush with the walls

Electronics bay under the removable strip: bus emulator 160 x 110, power supply 126 x 80

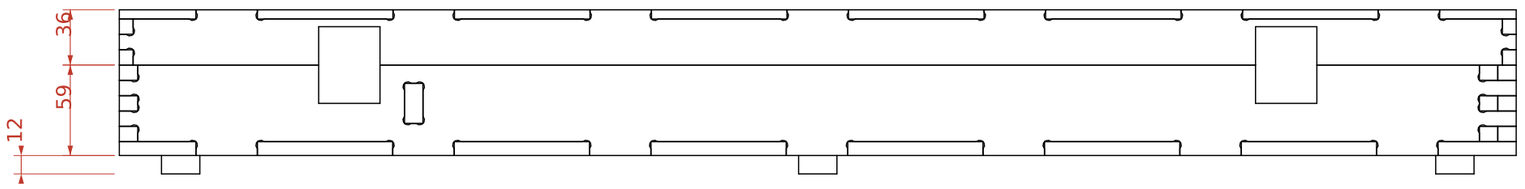
Rear plate (aluminium 2 mm): USB (option), DC IN, POWER, SUPPLY 2600/INT, MIDI IN, MIDI OUT

ARP 3620 replica - wooden case	Rev0.2 04.11.2026
Overview	sheet 1
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	renders not to scale

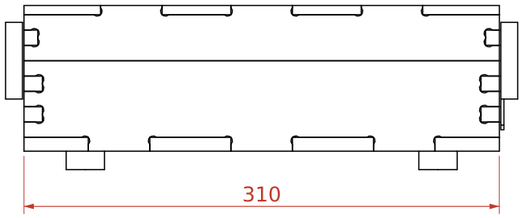
Top view, lid removed



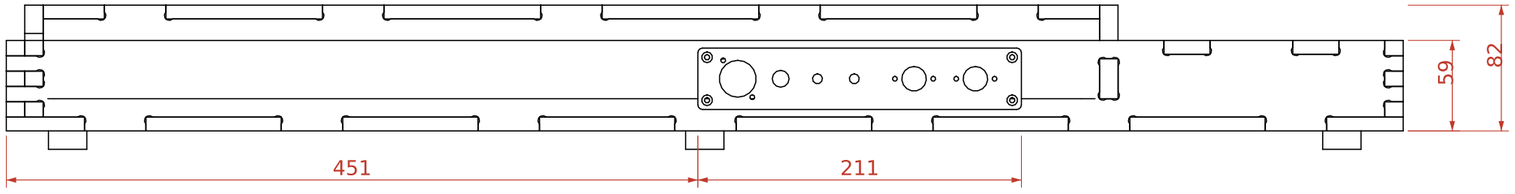
Front view with lid



Right side, with lid

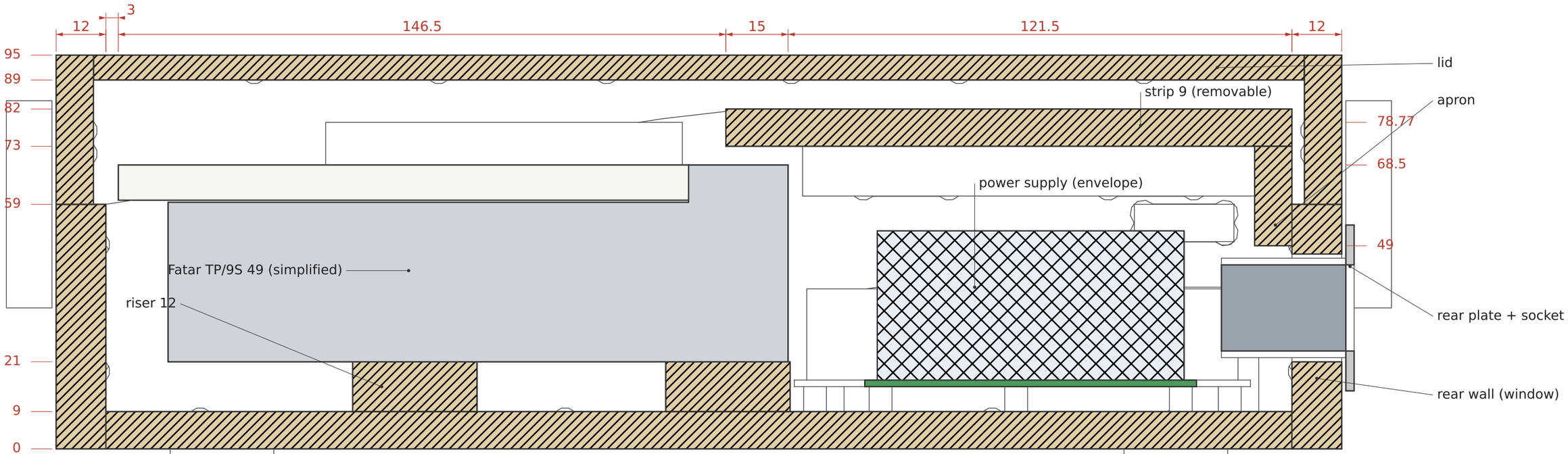


Rear view, lid removed

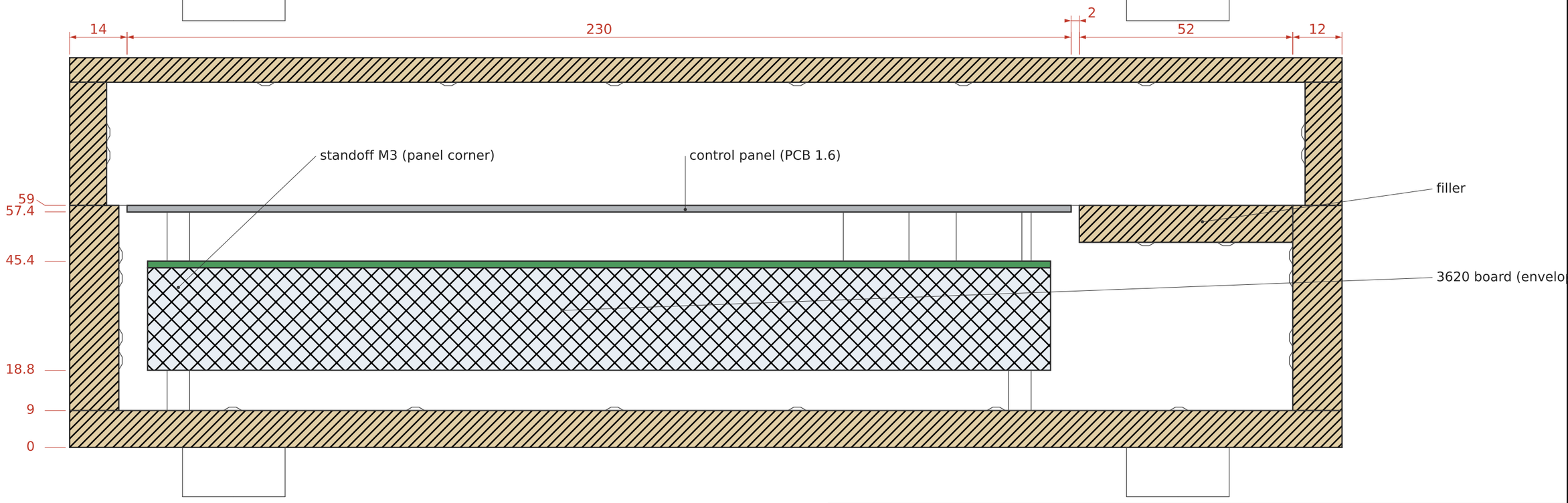


ARP 3620 replica - wooden case	Rev0.2 04.11.2026
General arrangement	sheet 2
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	scale 1:4

Section A-A at x = 440 (keyboard, power supply), seen from the right



Section B-B at x = 100 (control panel)

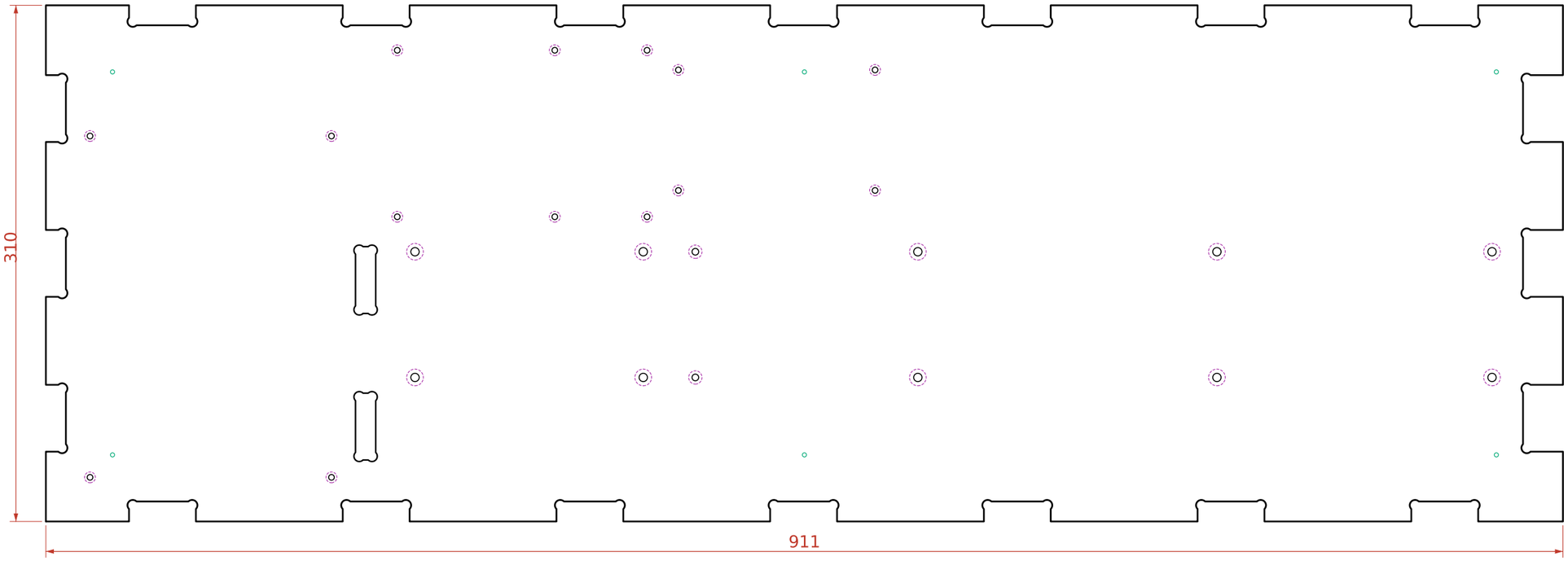


Heights follow from the keybed: the lower edge of the white key fronts (39 above the mounting plane, scaled from the Fatar drawing) sits 1 above the wall top. Measure it on the real keybed before cutting - all heights follow from it (params.py: KB_LIP). Board spacer 12 and 25 for parts and plugs under the 3620 board are estimates. Electronics are simplified boxes (parts height emulator 22, power supply 36).

ARP 3620 replica - wooden case	Rev0.2 04.11.2026
Sections	sheet 3
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	scale 1:1

Base - 1 x

birch plywood 9 mm, seen from above [base]



seen from above (inside). Countersinks: dashed = on the underside. Coordinates x/y from the front left corner.

dia 5 csk 10 far face (10 x) keybed, screw from below
221.65/86.5 358.72/86.5 523.61/86.5 703.2/86.5 868.4/86.5 221.65/162 358.72/162 523.61/162 703.2/162 868.4/162

dia 4 csk 8 far face (2 x) riser screw 4 x 20
390/86.5 390/162

dia 3.4 csk 6.5 far face (4 x) panel standoff M3
26.5/26.5 171.5/26.5 26.5/231.5 171.5/231.5

dia 3.4 csk 6.5 far face (6 x) emulator standoff M3
211/183 305.6/183 361/183 211/283 305.6/283 361/283

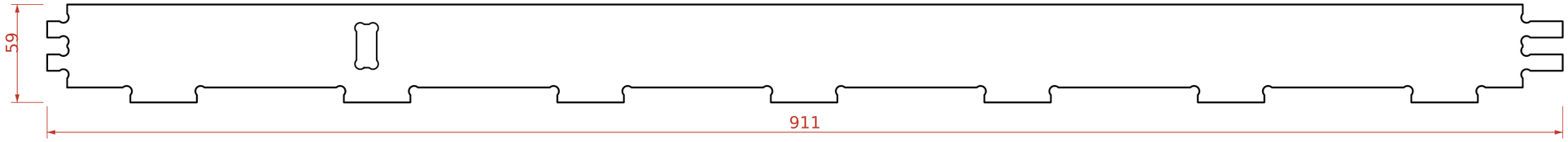
dia 3.4 csk 6.5 far face (4 x) power supply standoff M3
379.81/198.81 497.92/198.81 379.81/271.2 497.92/271.2

pilot 2.5 (6 x) foot
40/40 455.5/40 871/40 40/270 455.5/270 871/270

ARP 3620 replica - wooden case	Rev0.2 04.11.2026
Base	sheet 4
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	scale 1:3

Front wall - 1 x

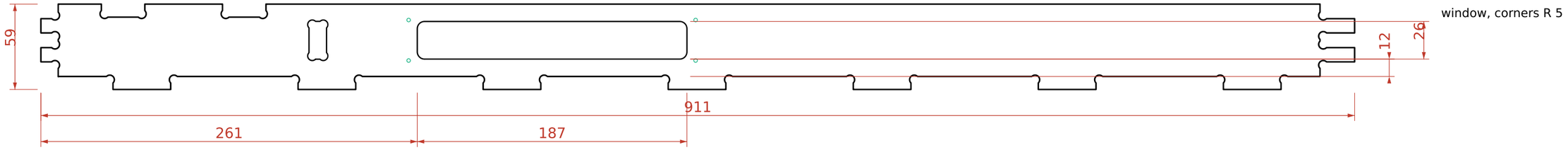
birch plywood 12 mm, seen from the front [front]



Seen from outside. Fingers at both ends, tabs into the base, slot for the left cheek. The right end takes end wall and right cheek together (24 deep).

Rear wall - 1 x

birch plywood 12 mm, seen from the front [rear]

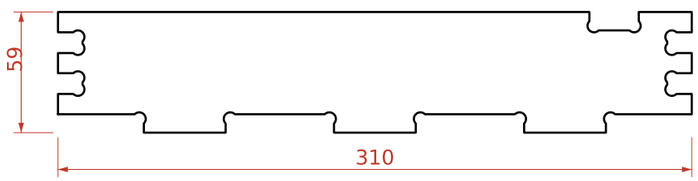


Seen from inside. Notches in the top edge for the filler tabs, slot for the left cheek, window for the bodies of the rear plate parts.

pilot 2.5 (4 x) rear plate, screw 3.5 x 16
255/20 454/20 255/48 454/48

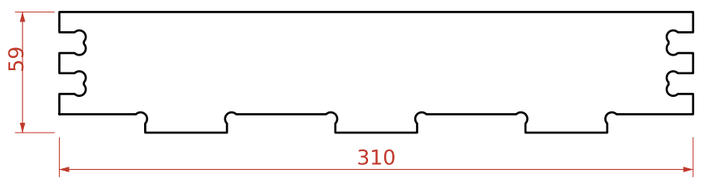
End wall left - 1 x

birch plywood 12 mm, seen from the right [end_l]



End wall right - 1 x

birch plywood 12 mm, seen from the right [end_r]

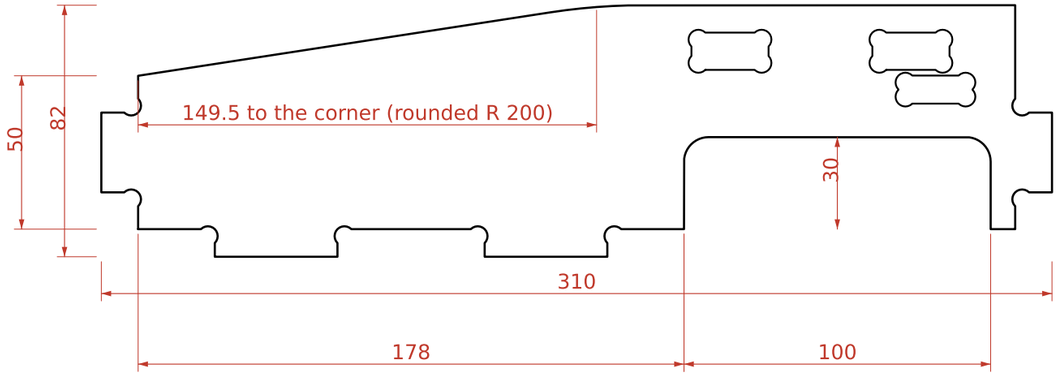


End walls seen from the right. Fingers at both ends (bottom, middle and top finger belong to the end wall), tabs into the base;
the left end wall has a notch in its top edge for the filler tab. Heights: 0 = top of the base, the tabs reach 9 below it.

ARP 3620 replica - wooden case	Rev0.2 04.11.2026
Walls	sheet 5
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	scale 1:3

Cheek left - 1 x

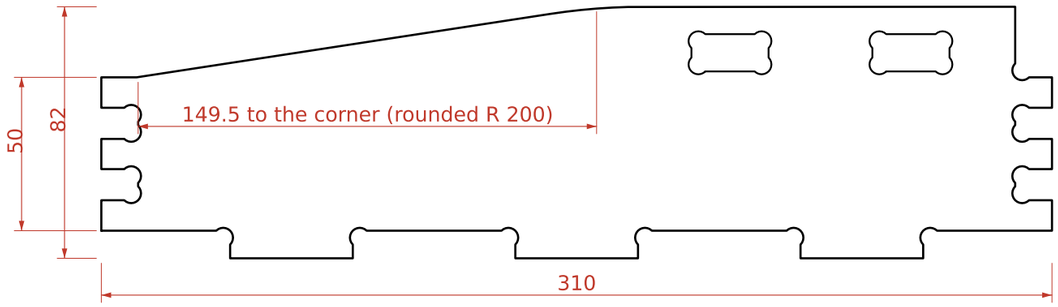
birch plywood 12 mm, seen from the right [cheek_l]



Seen from the right. u = 0 at the inner face of the front wall,
v = 0 at the top of the base; the tabs reach 9 below it.
Slope from 50 to 73 high, blending into the
flat top with R 200 (tangent points in the DXF).
Two slots for the strip cleats. Tolex on both faces and the top edge.
Tabs: front and rear through the walls, down through the base.
Slot at the rear for the filler tab.
Cable notch 100 x 30, corners R 8: cables
from the electronics bay to the panel section.

Cheek right - 1 x

birch plywood 12 mm, seen from the right [cheek_r]

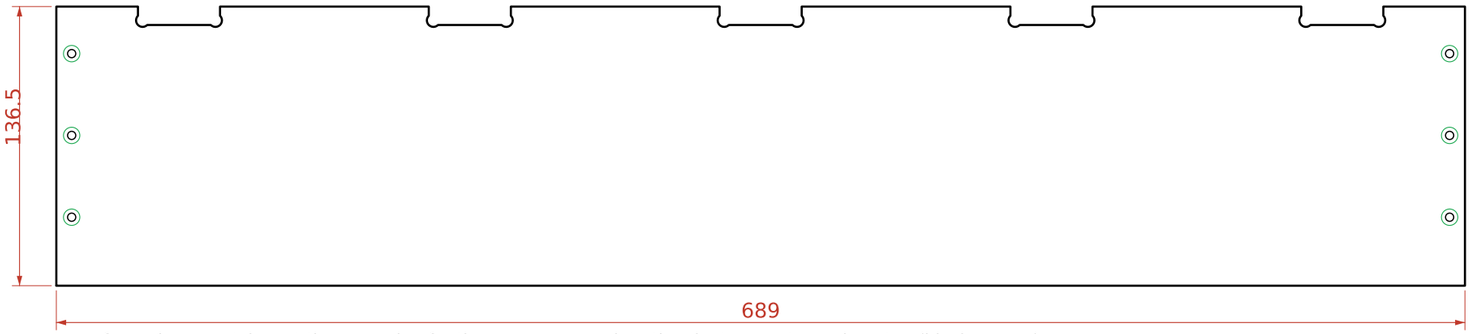


Seen from the right. u = 0 at the inner face of the front wall,
v = 0 at the top of the base; the tabs reach 9 below it.
Slope from 50 to 73 high, blending into the
flat top with R 200 (tangent points in the DXF).
Two slots for the strip cleats. Tolex on both faces and the top edge.
Stands against the right end wall and shares its joints:
same fingers into the front/rear wall, same tabs into the base.

ARP 3620 replica - wooden case	Rev0.2 04.11.2026
Cheeks	sheet 6
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	scale 1:2

Strip behind the keys - 1 x

birch plywood 9 mm, seen from above [strip]



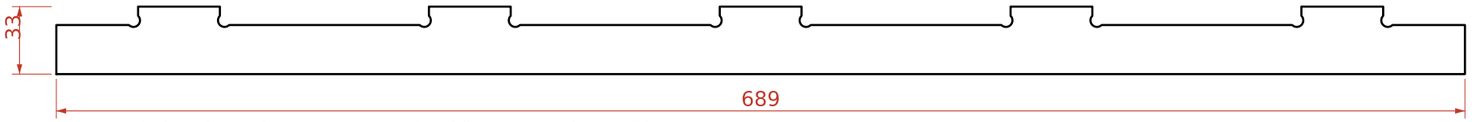
seen from above; notches at the rear edge for the apron; screws into the cleats, countersunk on top (black screws)

dia 4 csk 8 this face (6 x) screw 3.5 x 20 into cleat

7.5/33.5 681.5/33.5 7.5/73.5 681.5/73.5 7.5/113.5 681.5/113.5

Apron under the strip - 1 x

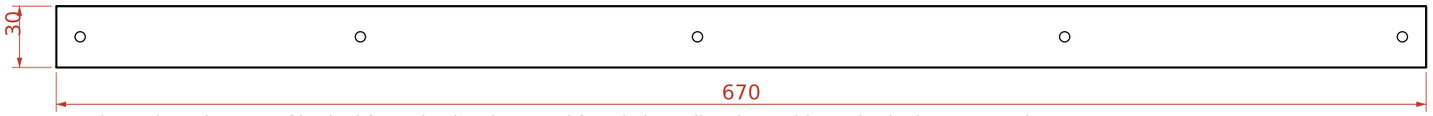
birch plywood 9 mm, seen from the front [apron]



tabs up into the strip notches, glued; reaches 10 mm below the wall top

Keybed riser - 2 x

birch plywood 12 mm, seen from above [riser_1, riser_2]



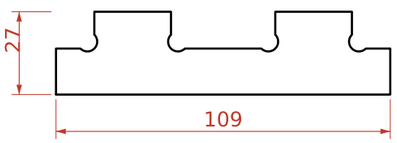
two risers along the rows of keybed feet, glued and screwed from below; align them with two keybed screws as pins

dia 5 (5 x) keybed screw

11.65/15 148.72/15 313.61/15 493.2/15 658.4/15

Strip cleat - 2 x

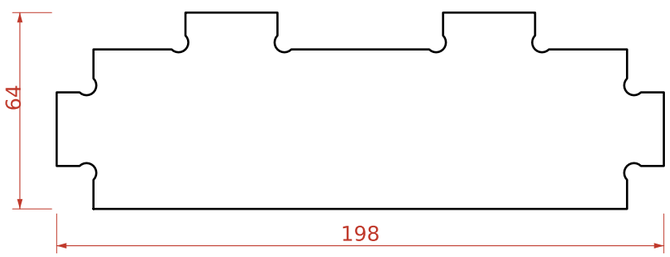
birch plywood 12 mm, seen from above [cleat_l, cleat_r]



lying flat, tabs through the cheek;
top 9 below the strip top

Filler behind the panel - 1 x

birch plywood 9 mm, seen from above [filler]

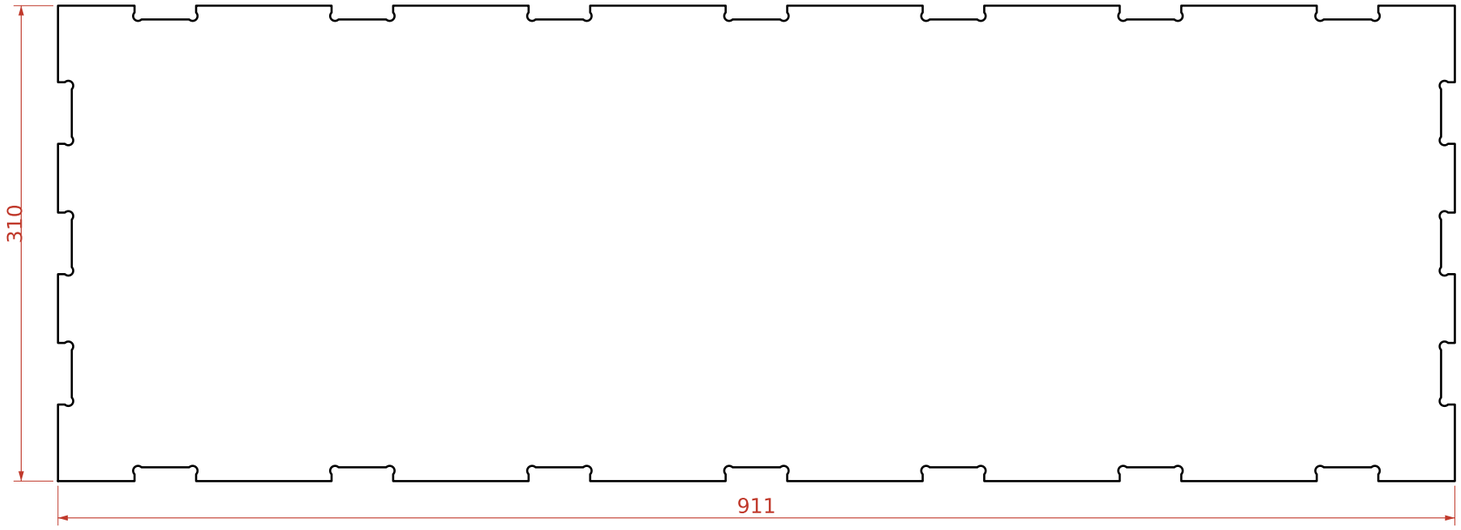


tabs into the left end wall, the rear wall and through the left cheek;
flush with the wall top

ARP 3620 replica - wooden case	Rev0.2 04.11.2026
Strip, risers, cleats, filler	sheet 7
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	scale 1:3 / 1:2

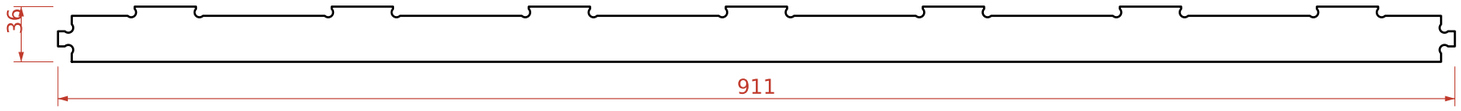
Lid top - 1 x

birch plywood 6 mm, seen from above [lid_top]



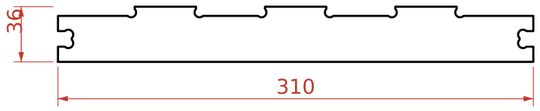
Lid front/rear - 2 x

birch plywood 9 mm, seen from the front [lid_front, lid_rear]



Lid end - 2 x

birch plywood 9 mm, seen from the right [lid_end_l, lid_end_r]

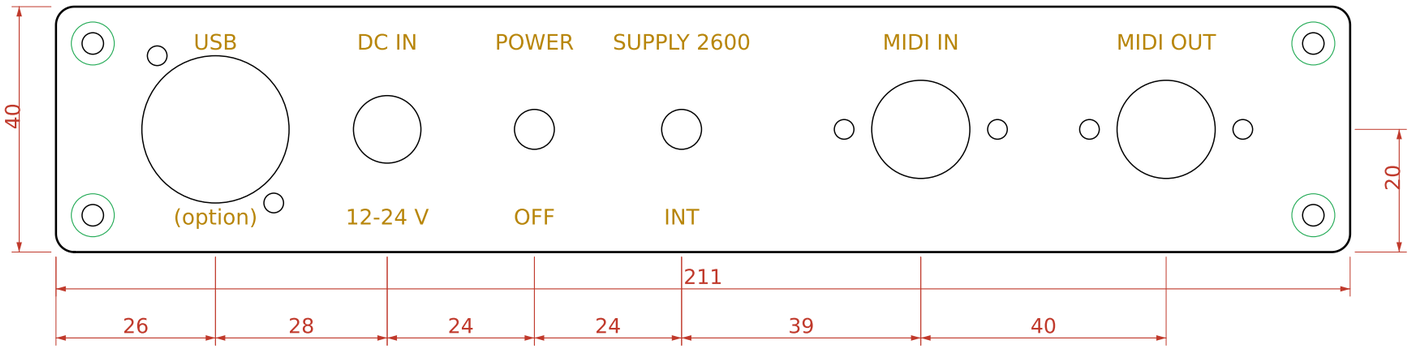


Walls 9 mm, inside height 30 above the case walls (7 clear above the strip).
Finger joints at the corners (the ends own the lower and the upper finger),
the wall tabs go into notches in the top. Glue; tolex outside and around the lower edge.
The lid sits on the case walls, flush outside. 4 latches centred on the joint (2 front, 2 rear)
keep it in place; the latches also locate it sideways.

ARP 3620 replica - wooden case	Rev0.2 04.11.2026
Lid	sheet 8
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	scale 1:4

Rear plate - 1 x

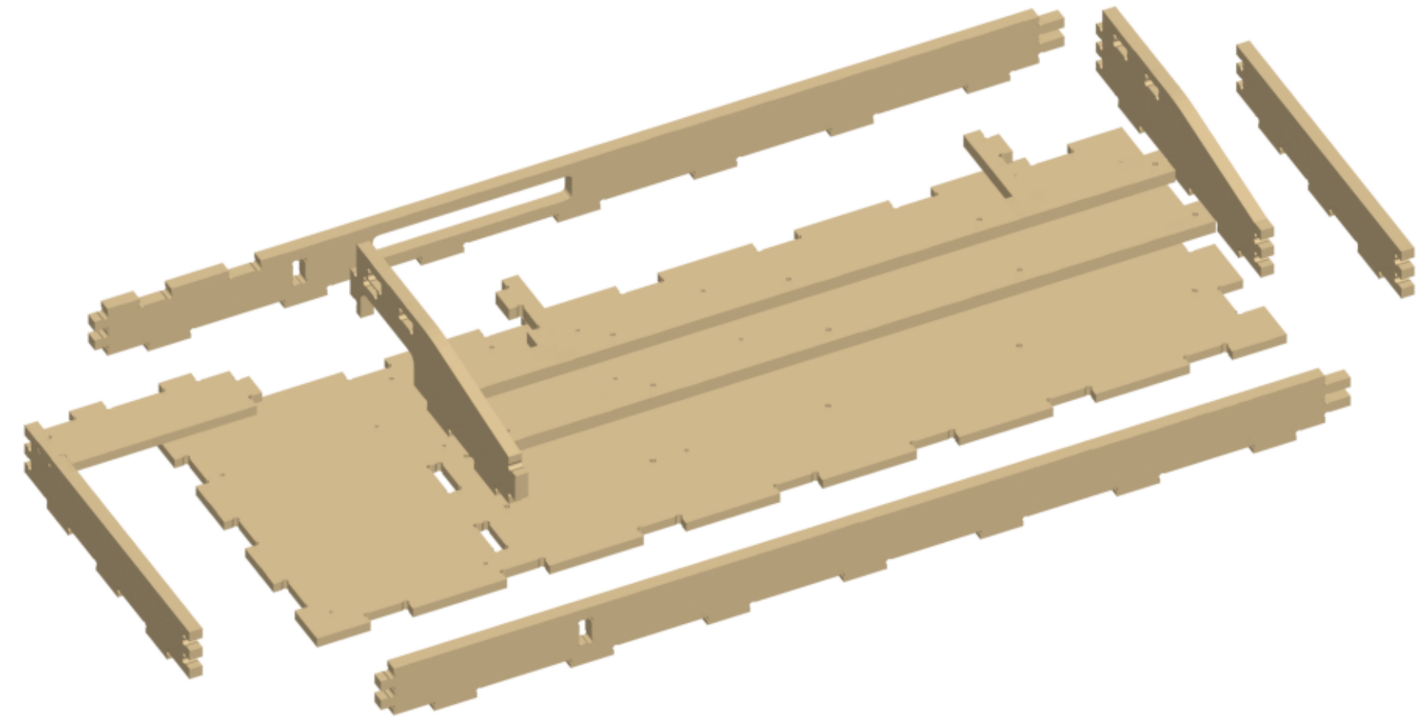
aluminium 2 mm, seen from the rear [plate]



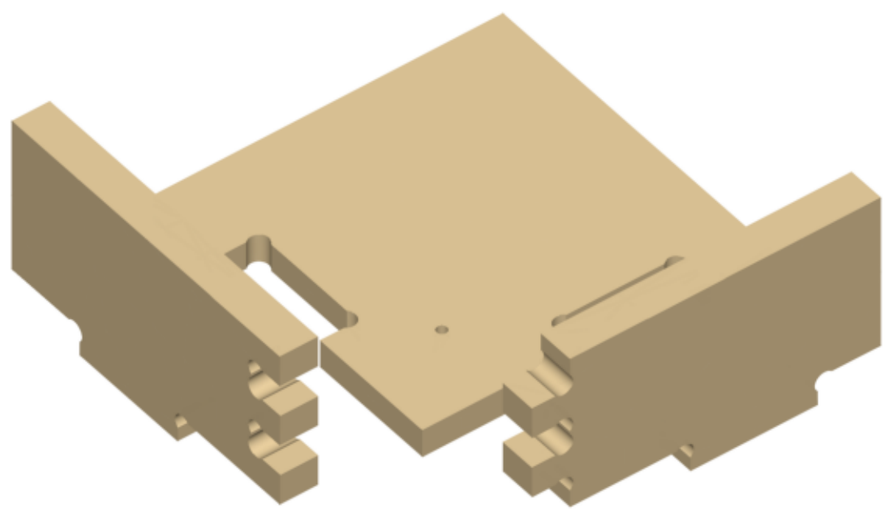
dia 24 (1 x) USB
 26/20
dia 16 (1 x) MIDI IN
 141/20
dia 16 (1 x) MIDI OUT
 181/20
dia 11 (1 x) DC IN
 54/20
dia 6.5 (1 x) POWER
 78/20
dia 6.5 (1 x) SUPPLY 2600
 102/20
dia 3.5 csk 7 this face (4 x) screw 3.5 x 16
 6/6 205/6 6/34 205/34
dia 3.2 (2 x) USB
 35.5/8 16.5/32
dia 3.2 (2 x) MIDI IN
 128.5/20 153.5/20
dia 3.2 (2 x) MIDI OUT
 168.5/20 193.5/20

Aluminium 2 mm, black anodised, legend engraved or printed white.
Seen from outside (from the rear of the case).
Mounted on the outside of the rear wall, over the tolex,
with 4 countersunk screws 3.5 x 16 (pilot holes in the rear wall).
Seen from the rear, the plate starts 451 from the left end of the case;
its lower edge is 14 above the underside of the base.
The window in the rear wall takes the bodies of the parts.
Behind the emulator only 25 mm depth is free; the deep USB
extension sits at the left end, behind the power supply.
Hole sizes are placeholders until the parts are chosen.

ARP 3620 replica - wooden case	Rev0.2 04.11.2026
Rear plate	sheet 9
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	scale 1:1



plugged together: finger joints, tabs and slots (exploded)

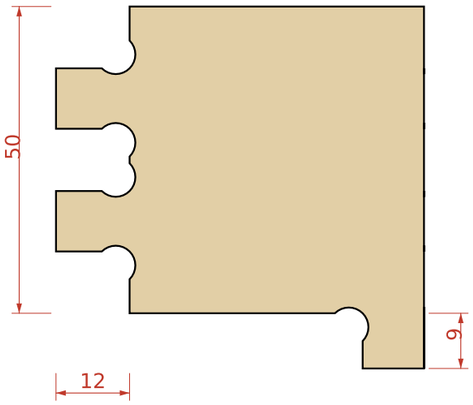


front left corner, parts pulled apart

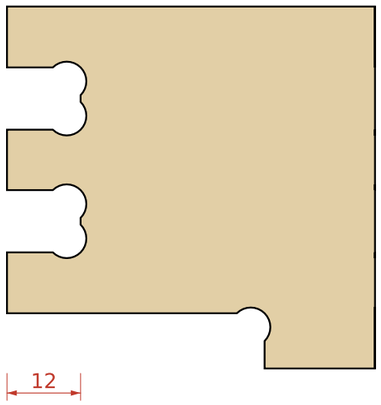
Assembly order (carcass)

1	Base	
2	Keybed riser	down
3	Keybed riser	down
4	Cheek left	down
5	Cheek right	down
6	Strip cleat	to the left
7	Strip cleat	to the right
8	Filler behind the panel	to the right
9	Front wall	to the rear
10	Rear wall	to the front
11	End wall left	to the right
12	End wall right	to the left

front wall, left end (seen from outside)



left end wall, front end (seen from inside)



Dogbones R 3.2 in every inside corner (end mill up to 6.35 mm).
5 fingers over 50 mm: bottom, middle and top finger on the end wall.
Play per joint: notches and slots 0.15 mm wider than the tab
(split over both parts at the fingers). Joint depth = sheet thickness:
measure the plywood and enter it before cutting (params.py, T_*).
Tab ends sit flush with the outside, under the tolex.

Joints in the 3D model: no part overlaps another, every joint touches,
each part slides into place in the order below (model/check.txt).

ARP 3620 replica - wooden case	Rev0.2 04.11.2026
Joints, CNC, assembly order	sheet 10
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	detail 1:1, renders not to scale

Cut list (finished sizes, mm)

part	qty	material	length	width	covering / finish
Base	1	birch plywood 9	911	310	underside painted black, edges covered by the tolex of the walls
Front wall	1	birch plywood 12	911	59	tolex outside, over the top edge, 20 mm down the inside
Rear wall	1	birch plywood 12	911	59	tolex outside, over the top edge, 20 mm down the inside
End wall left	1	birch plywood 12	310	59	tolex outside, over the top edge
End wall right	1	birch plywood 12	310	59	tolex outside, over the top edge
Cheek left	1	birch plywood 12	310	82	tolex on both faces and the top edge
Cheek right	1	birch plywood 12	310	82	tolex on both faces and the top edge
Keybed riser	2	birch plywood 12	670	30	black paint
Strip behind the keys	1	birch plywood 9	689	136.5	tolex on top, front edge and ends
Apron under the strip	1	birch plywood 9	689	33	black paint, glued into the strip
Strip cleat	2	birch plywood 12	109	27	black paint
Filler behind the panel	1	birch plywood 9	198	64	tolex on top and front edge
Lid top	1	birch plywood 6	911	310	tolex outside
Lid front/rear	2	birch plywood 9	911	36	tolex outside and over the lower edge
Lid end	2	birch plywood 9	310	36	tolex outside and over the lower edge
Rear plate	1	aluminium 2	211	40	black anodised, engraved white (or printed)

net area: birch plywood 9 mm: 0.50 m2, birch plywood 12 mm: 0.24 m2, birch plywood 6 mm: 0.28 m2, aluminium 2 mm: 0.01 m2

Hardware

- Tolex, black (vinyl case covering), approx. 1.1 m2 incl. 25 % waste
  outside, see covering notes
- contact cement for tolex, wood glue (PVAc), wood filler
  all joints glued; filler for dogbones and joint lines under the tolex
- 2 x countersunk wood screw 4 x 20
  base -> risers (from below)
- 6 x countersunk wood screw 3.5 x 20, black
  strip -> cleats (removable strip)
- 4 x countersunk screw 3.5 x 16, black
  rear plate
- 10 x keybed screw from below through base and riser
  length ~ 21 mm + thread in the keybed foot; type to be confirmed on the keybed (hole 4.5)
- 4 x M3 standoff, total length 48.4 mm (e.g. 40 + 8 and a washer)
  panel corners (panel PCB 1.6), M3 x 12 countersunk from below, M3 x 6 black from the top
- 10 x M3 standoff 6 mm, 10 x M3 x 12 countersunk
  emulator (6), power supply (4)
- 6 x rubber foot dia 25 x 12 with screw
- 4 x draw latch (case latch), small, ~40 mm wide
  lid: 2 front, 2 rear, centre on the lid joint
- rear plate parts: 2 x DIN 5-pin panel socket, DC jack 5.5/2.1, 2 x toggle switch, USB extension (option)
  hole sizes on the plate are placeholders - confirm with the chosen parts
- optional: 8 x case corner, carrying handle

Assembly

1. CNC: cut all parts as per DXF, drill and countersink (CSK_BACK = countersink on the far face).
2. Check KB_LIP and the screws on the real keybed first; fit the keybed onto the risers dry.
3. Dry fit the whole case first (order on sheet 10), then take it apart and glue.
4. base on the bench, inside up
5. risers: glue, screw from below, keybed holes aligned
6. left cheek: tabs down into the slots of the base
7. right cheek: tabs down into the notches at the right edge of the base
8. strip cleats: from the inside through the cheeks
9. filler: right tab into the left cheek, hold it level
10. front wall: over the cheek tabs, tabs into the base
11. rear wall: over the cheek tabs and the filler tabs
12. end walls from the sides: fingers, base tabs, filler tab
13. Strip unit: apron tabs up into the strip, glued; the strip stays removable (screws into the cleats).
14. Lid: front and rear onto the top, ends from the sides; glue.
15. Fill dogbones and joint lines on the outside, sand flush, round all outside edges to about R2.
16. Paint the inside black, then tolex outside with contact cement; wrap 20 mm over the top edges.
17. Rear plate, latches, feet; panel on its standoffs, electronics on the base, keybed screwed from below.

Open points

- Sheet thickness: measure the plywood, enter it in params.py (T_*) and regenerate.
- KB_LIP 39 (key front lower edge) scaled from the Fatar drawing - measure.
- Keybed screw type and length (foot hole 4.5).
- Board spacer 12 and parts height under the 3620 board.
- Rear plate parts and their hole sizes.
- Latch type; optional handle and case corners.

ARP 3620 replica - wooden case	Rev0.2 04.11.2026
Cut list, hardware, assembly	sheet 11
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	mm